

MD. TAKOAT HOSSAIN

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EEG-based Mental Health Diagnostics • Graph Neural Networks for Neural Signal Processing • Brain-Computer Interfaces
• Affective Computing • Biomedical AI

EDUCATION

B.Sc. in Electrical and Electronic Engineering

2022- 2026

Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh

CGPA: 3.70 / 4.00 (Top 5%; Unofficial Result) • H.S.C. GPA: 5.00 / 5.00 (2020) • S.S.C. GPA: 5.00 / 5.00 (2018)

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher: SUST EEE Department

MindMap: An EEG and AI-Driven Study for Mental Health Assessment and Categorization | Supervisor: Dr. Md. Rasedujjaman

- Designed a Spatial-Temporal Graph Attention Network (ST-GAT) treating 62 EEG electrodes as graph nodes to model distributed neural connectivity patterns underlying Major Depressive Disorder (MDD)
- Implemented cross-domain transfer learning: pre-trained on SEED-IV (4-class emotion, 46.30% accuracy), fine-tuned on MODMA clinical dataset (MDD vs. HC, 67.57% accuracy)
- Validated framework on independent cohort of 294 subjects- achieved 71.79% diagnostic accuracy (AUC = 0.65) with statistical significance $p < 0.001$
- Identified Frontal-Midline coherence (FP1–Fz, Delta/Theta bands) and Parietal Alpha (PO8, $p = 0.004$) as key neurological biomarkers of depression via attention-weight analysis
- Conducted interpretability analysis through graph attention visualization; results align with established neurobiological theories of prefrontal dysregulation in MDD

Research Project: Affective Computing / Speech Processing

2024

BSER: Bengali Speech Emotion Recognition using MFCC Features

- Compared ML classifiers (SVM, RF, MLP) on MFCC features extracted from two Bengali speech datasets (SUBESCO, BanglaSER) for emotion recognition
- Resulted in peer-reviewed conference publication at ICERIE 2025 (Atlantis Press)

PUBLICATIONS

[1] Hossain, M. T., et al. (2025). BSER – Bengali Speech Emotion Recognition using MFCC Features Based on SUBESCO and BanglaSER: A Comparative Study of Machine Learning Models. In Proceedings of ICERIE 2025, Atlantis Press. DOI: [10.2991/978-94-6463-884-4_77](https://doi.org/10.2991/978-94-6463-884-4_77)

TECHNICAL SKILLS

Programming: Python (primary), MATLAB, C/C++, Verilog

Deep Learning: Graph Attention Networks (ST-GAT), Transfer Learning, Supervised/Semi-supervised Models, PyTorch / TensorFlow

Signal Processing: EEG preprocessing, frequency-band decomposition (Delta/Theta/Alpha/Beta/Gamma), FFT, spectral analysis, digital filtering, artifact removal

Datasets & Tools: SEED-IV, MODMA, Zenodo EEG Depression Dataset; dataset preprocessing pipelines, cross-domain evaluation

Hardware / VLSI: Cadence Virtuoso, Silvaco SPICE, ModelSim, LTSpice, Multisim, Proteus

Other Software: PowerWorld Simulator, CST Studio, AutoCAD

ADDITIONAL PROJECTS

Noise and Pitch Detection of Audio Samples

2024

Designed signal processing pipeline for pitch estimation and noise analysis using FFT-based spectral decomposition.

Electric Vehicle Prototype System- SUST AUTO-Drive

2023

Led electrical architecture design for EV prototype; responsible for motor control and power distribution subsystems.

ACHIEVEMENTS / AWARDS

- University Merit Scholarship, Shahjalal University of Science and Technology (2021)- awarded a merit-based scholarship of BDT 1000 in recognition of outstanding academic performance (CGPA 3.77) in the 1st semester.
- University Merit Scholarship, Shahjalal University of Science and Technology (2021)- awarded a merit-based scholarship of BDT 2,400 in recognition of outstanding academic performance (CGPA 3.64) in the 2nd year.

REFERENCES

Dr. Md. Rasedujaman- Associate Professor, Dept. of EEE, SUST • mrased-eee@sust.edu (Thesis Supervisor)

Md. Shariful Islam- Assistant Professor, Dept. of EEE, SUST • shariful-eee@sust.edu (Thesis Co-Supervisor)

Dr. Mohammad Kamruzzaman Khan Prince- Associate Professor, Dept. of EEE, SUST • kamruzzamaneee@sust.edu